Marketing ROI & Budget Reallocation — Decision Memo

1. Goal & KPIs + Attribution Rule

The primary objective of this analysis is to optimize marketing spend allocation across channels to maximize revenue and improve overall return on investment (ROI).

Primary KPIs

- Sessions (traffic volume)
- Orders (conversions)
- Revenue
- Conversion Rate (CVR = Orders / Sessions)
- Return on Ad Spend (ROAS = Revenue / Spend)
- Cost per Acquisition (CPA = Spend / Orders)

Attribution Rule

A **last-touch attribution model** is used, where the final channel before conversion receives full credit for the revenue.

This helps understand:

- Which channels directly drive conversions
- Immediate revenue contribution by channel

2. Data & Methodology (ETL + Definitions)

Data Sources

- sessions.csv → user sessions, device, channel
- orders.csv → order-level data
- order_items.csv → product-level revenue
- campaigns.csv → campaign details
- products.json → product categories

ETL Process

- Merged sessions with orders using `session_id`
- Joined order_items with products to get category
- Aggregated:
    - Revenue
    - Orders
    - Sessions
    -

- Created derived metrics:
  - ROAS = Revenue / Spend
  - CAC = Spend / Orders
  - Conversion Rate = Orders / Sessions

Key Definitions

- New User: First-time visitor
- Returning User: Repeat visitor
- Channel: Marketing source (Search, Email, Paid Social, etc.)
- Category: Product grouping (Books, Fashion, etc.)

3. Key Insights

1. Search channel dominates revenue contribution, making it the most impactful marketing channel.
2. Email shows high efficiency — lower spend but strong ROAS, indicating cost-effective performance.
3. Paid Social has high CAC, meaning it is relatively expensive to acquire customers.
4. New users have slightly higher conversion rates, suggesting strong acquisition strategies.
5. Returning users generate stable revenue per session, indicating good retention quality.
6. Mobile contributes a significant share of traffic and revenue, but performance varies across channels.
7. Some categories (like Fashion & Books) consistently perform well across channels, indicating strong product-market fit.

4. Attribution Findings (What Looks Good / Bad)

What Looks Good

- Search contributes the highest share of attributed revenue
- Email performs efficiently with strong returns
- Organic traffic supports consistent performance

What Looks Weak

- Paid Social shows lower efficiency despite high spend
- Referral contributes low revenue share
- Some channels show diminishing returns with higher spend

5. Regression Findings (Estimated Impact + Limitations)

Findings

- Search has the highest marginal impact on revenue
- Paid Social and Email show moderate positive impact
- Organic and Referral show lower or negative coefficients

Interpretation

- Channels with higher coefficients → stronger revenue drivers
- Negative coefficients may indicate:
  - Multicollinearity
  - Attribution overlap
  - External factors not captured in model

Limitations

- Model has relatively **low R^2**, meaning:
  - Not all revenue variation is explained
- Results should be treated as:
  Directional insights, not exact causal conclusions

6. Budget Plan + 30-Day Impact Estimate

Proposed Budget Reallocation

Channel	**Current %**	**New %**	**Action**
Search	46%	40%	Slight decrease
Paid Social	38%	20%	Significant reduction
Email	5%	15%	Increase
Organic	1%	10%	Increase
Referral	9%	15%	Increase

Strategy

- Reduce spend from low-efficiency channels (Paid Social)
- Increase investment in:
  - Email (high ROI)
  - Organic (long-term growth)
  - Referral (scalable potential)

30-Day Impact Estimate

- Base Case:
  +8% to +12% revenue increase
- Best Case:
  +15% to +20% revenue increase
- Worst Case:
  +3% to +5% revenue increase

Assumptions

- Channel performance remains stable
- No major seasonality impact
- User behavior remains consistent

7. Risks / Limitations + Next Steps

Risks

- Attribution model may oversimplify customer journey
- Regression results affected by limited variables
- External factors (seasonality, competition) not included

Next Steps

1. Run A/B tests on budget allocation changes
2. Improve attribution model (multi-touch attribution)
3. Collect more granular campaign-level data
4. Monitor performance weekly and adjust dynamically
5. Optimize Paid Social campaigns instead of full reduction

Conclusion

This analysis provides a data-driven approach to optimize marketing spend.By reallocating budget toward high-performing channels, the business can:

Increase revenue

Improve efficiency

Reduce acquisition costs